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**TEAM NAME:** Infinity Loopers

**COLLEGE NAME:** Sri Venkateswara College of Engineering

**PROBLEM STATEMENT:** Trustshield: AI Platform For Deepfake and Digital Abuse Detection

# PROBLEM STATEMENT

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## Problem Statement

In today's world, the rapid advancement of generative AI tools such as OpenAI models and DeepFaceLab has made it easy to create highly realistic deepfake images, videos, and cloned voices. These synthetic media technologies are increasingly misused for fraud, identity impersonation, misinformation, deepfake sexual abuse content, non-consensual image sharing, voice cloning scams, and bypassing video KYC or banking verification systems. This misuse creates serious risks to digital safety, trust, and personal privacy.

## Existing Solutions

- Most deepfake detection tools only give a simple **"real or fake"** result, which does not explain the manipulation clearly.
- Many systems work only on specific platforms, so they cannot protect users across different apps or websites.
- Some tools are slow and not designed for real-time detection, allowing fake content to spread quickly.
- Existing solutions often struggle to detect complex manipulations involving images, videos, and audio together.
- Many systems do not provide clear explanations or evidence about how the media was manipulated.

## Issues

- Rapid rise of deepfake content enabling identity impersonation and digital fraud.
- Non-consensual image sharing and deepfake abuse causing emotional and reputational harm.
- Voice cloning scams and AI-generated media being used for financial fraud.
- Deepfakes bypassing video KYC and identity verification systems, creating security risks.
- Lack of reliable real-time detection tools for identifying manipulated images, videos, and audio.

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# SOLUTION

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## 1.MEDIA CAPTURE & BROWSER EXTENSION INTERFACE

**Purpose**: Captures images, videos, and audio from social media platforms and sends them to the system for real-time deepfake analysis.

**Implementation**: A JavaScript browser extension captures webpage media and sends it via REST APIs to a Python backend (Flask/FastAPI) for real-time deepfake analysis, with Flutter enabling mobile uploads.

## 2.AI DEEPFAKE DETECTION ENGINE

**Purpose:** Detects whether images, videos, or audio are AI-generated or manipulated using deepfake technologies.

**Implementation:**Captured media is analyzed using deep learning models (CNN) built with PyTorch/TensorFlow, using OpenCV for video frame analysis & Librosa spectrograms for audio, to classify content as real, suspicious, or deepfake.

## 3.AI DIGITAL DNA FINGERPRINTING & WATERMARK DETECTION

**Purpose**: Identifies hidden AI-generation signatures and detects invisible watermarks in synthetic media.

**Implementation:** Uses image forensic analysis with OpenCV and Scikit-image to detect GAN/diffusion artifacts and trained watermark detection models to identify AI-generated patterns.

## 4.SPAM, SCAM & DIGITAL ABUSE DETECTION

**Purpose**: Detects scam messages, fraudulent media, and abusive synthetic content to prevent digital harassment and misinformation.

**Implementation:** Uses NLP models (Hugging Face Transformers) to analyze text for scams/phishing and computer vision models in PyTorch to detect harmful or manipulated media.

## 5.REVERSE DEEPFAKE SEARCH & COMMUNITY INTELLIGENCE

**Purpose**: Identifies whether similar deepfake media has appeared online and alerts users about possible misinformation campaigns.

**Implementation**: Uses perceptual hashing (pHash) and FAISS vector similarity search to compare media with a database and detect previously reported deepfakes.

## 6.EXPLAINABLE AI REPORTING & USER ALERT SYSTEM

**Purpose**: Provides clear explanations and evidence showing why a piece of media is flagged as manipulated instead of simply labeling it as fake.

**Implementation:** Uses Explainable AI techniques like Grad-CAM (PyTorch) and OpenCV frame analysis to generate manipulation heatmaps and display alerts through the extension interface.

# FLOW DIAGRAM

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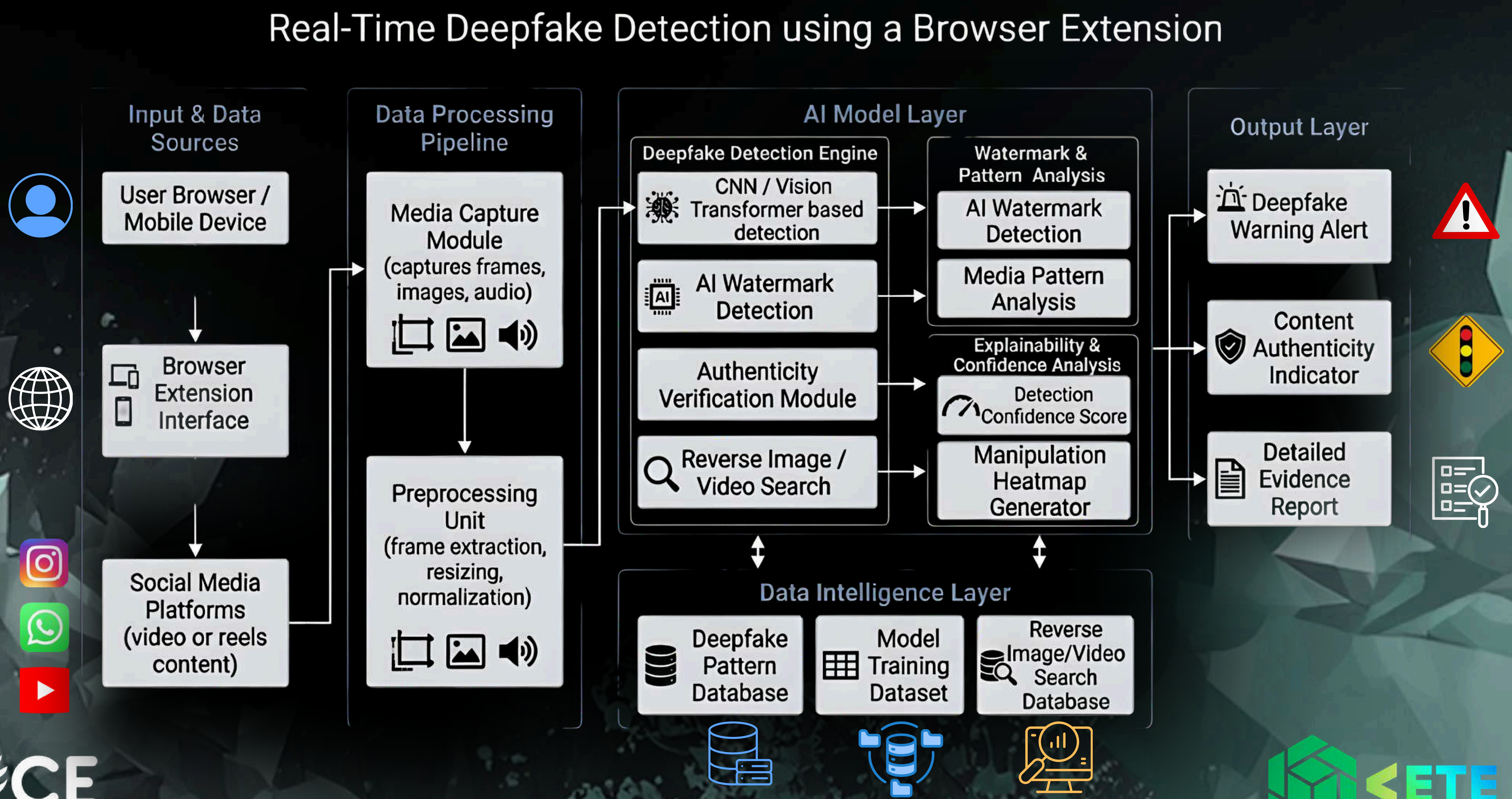


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# TECHNICAL REQUIREMENTS

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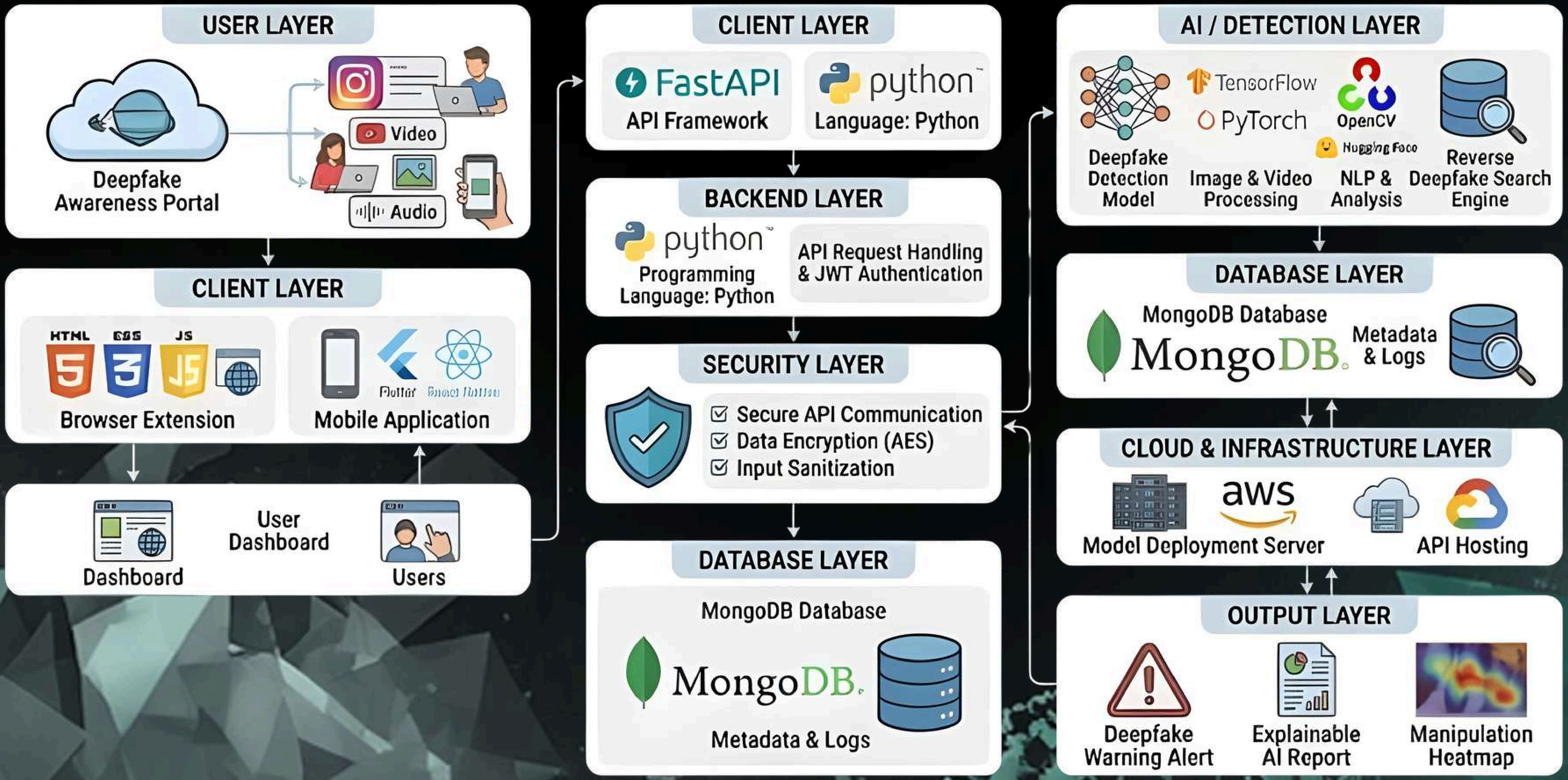


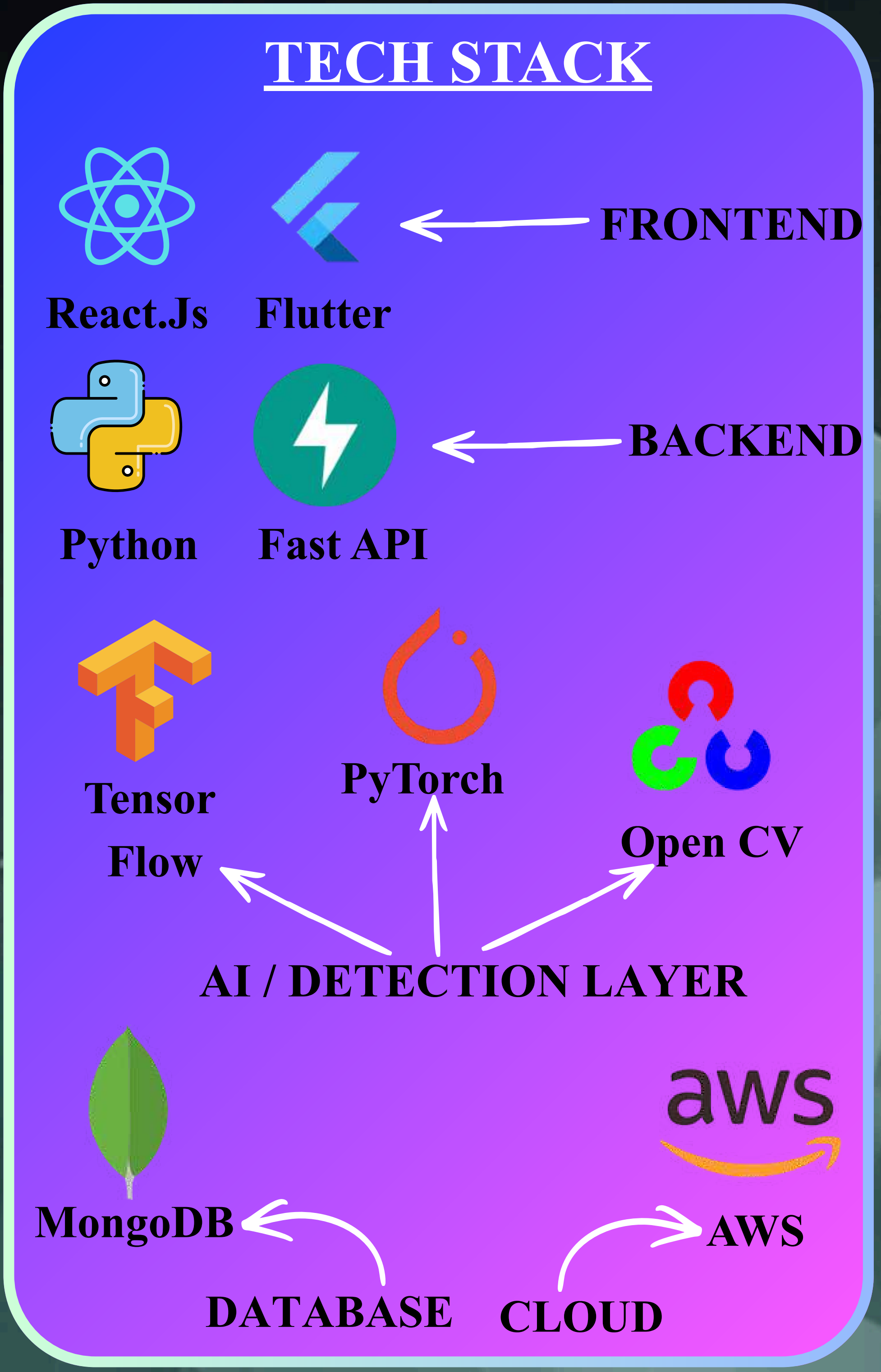


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# NOVELTY / USE CASE

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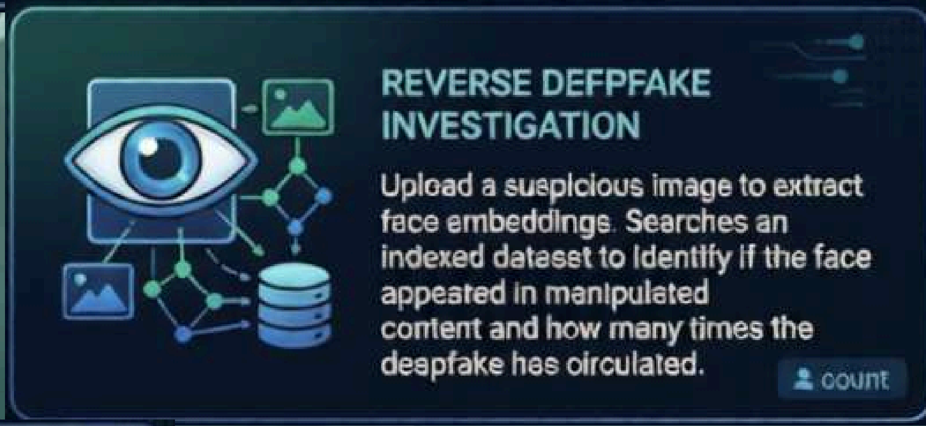
## NOVELTY

| Features | Our System Novelty |
|---|---|
| Real-Time Browser Extension Detection | Instead of uploading media to a website, the system works as a browser extension.<br>It detects deepfake or AI-generated media while users browse social media platforms (e.g., Instagram videos or reels). |
| Forensic Explainable AI Report | Existing tools only return a probability score.<br>Our system generates a detailed forensic analysis report, including:<br>Manipulation heatmap<br>Audio-video mismatch graphs |
| Reverse Deepfake Search/Backtracking | Our system backtracks and finds from where the deepfake content has orginated |
| Abusive Content Detection | Our system includes AI-based abusive content detection that analyzes:<br>Harmful text or captions<br>Manipulated faces used in harassment<br>Scam or misleading speech patterns |
| Community Deepfake Alert Network | If multiple users detect similar deepfake patterns, the system:<br>identifies coordinated manipulation campaigns<br>learns trending deepfake signatures<br>alerts other users about potential threats. |

## USE CASE







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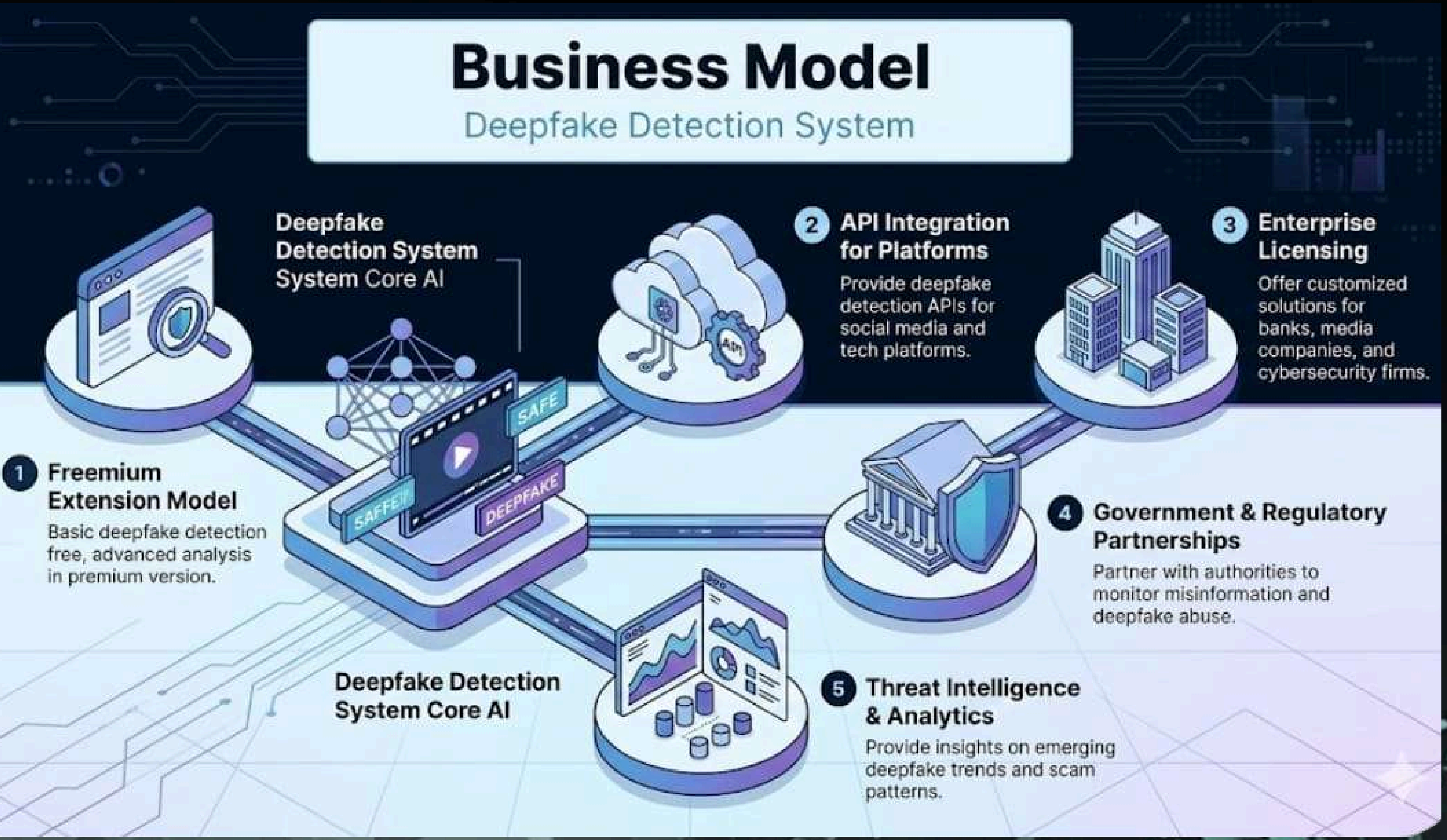
# BUSINESS MODEL / TARGET MARKET




## Target Market

**1. Social Media Users**
Individuals who want to verify whether images, videos, or reels are AI-generated or manipulated while browsing platforms like Instagram, YouTube, and TikTok.

**2. Journalists and Fact-Checking Organizations**
Media professionals who need tools to verify the authenticity of digital media before publishing news (e.g., organizations like BBC and Reuters).

**3. Social Media Platforms**
Technology companies that need automated systems to detect deepfakes, AI-generated media, and misinformation across their platforms (such as Meta Platforms).

**4. Cybersecurity and Government Agencies**
Cybercrime units and digital forensic investigators who analyze manipulated media used in scams, impersonation attacks, or misinformation campaigns (for example, Indian Computer Emergency Response Team).

**5. Businesses and Enterprises**
Organizations that want protection against deepfake fraud, impersonation attacks, and AI-generated misinformation targeting their brand or executives.

# TEAM DETAILS

INFINITY LOOPERS

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| Team | Name | Branch / Stream | Year of Study | College Name | Email | Contact Number |
|---|---|---|---|---|---|---|
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